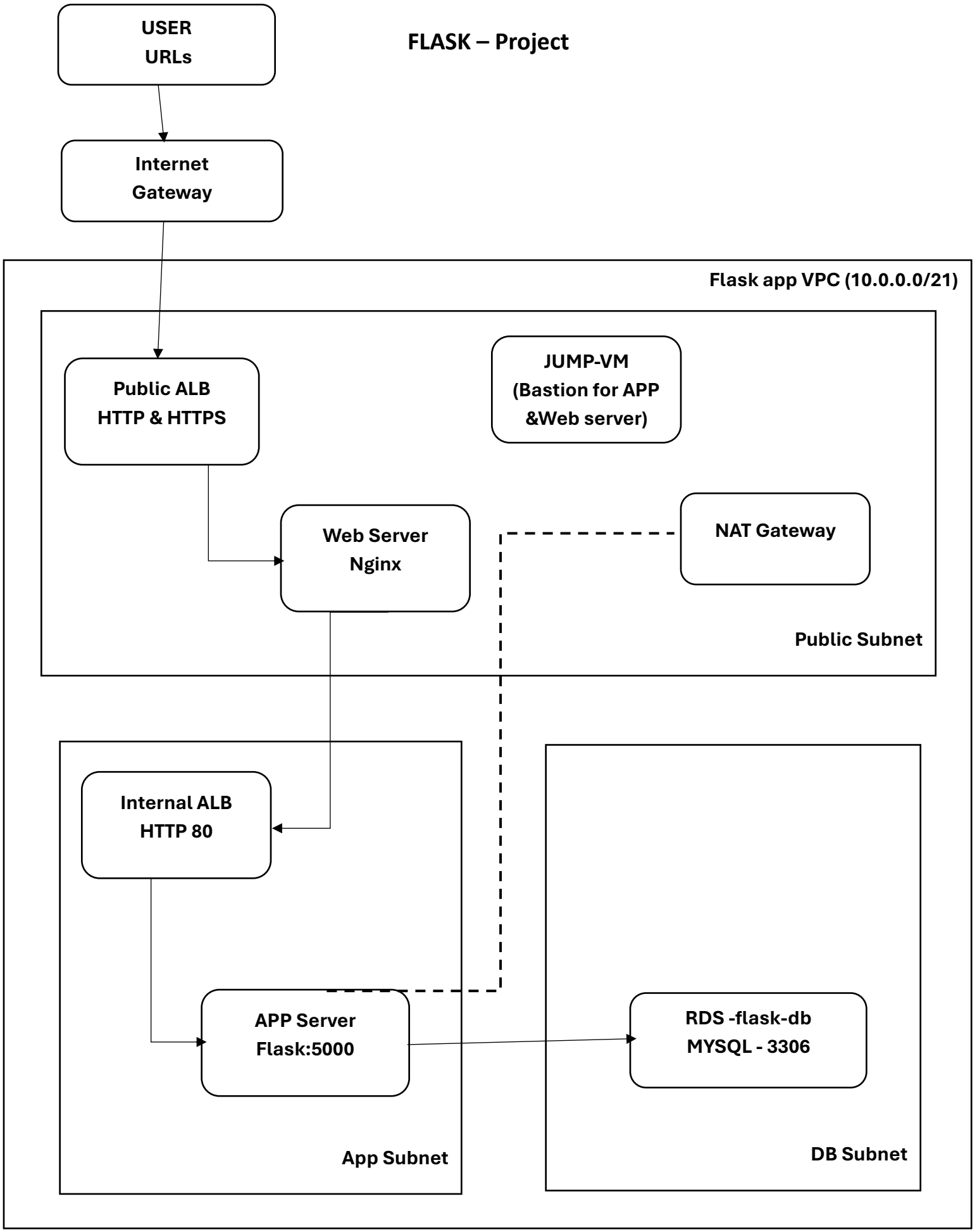

USER
URLs
FLASK – Project
Internet
Gateway
Flask app VPC (10.0.0.0/21)
Public ALB
HTTP & HTTPS
JUMP-VM
(Bastion for APP
&Web server)
Web Server
Nginx
NAT Gateway
Public Subnet
Internal ALB
HTTP 80
APP Server
Flask:5000
RDS -flask-db
MYSQL - 3306
App Subnet
DB Subnet

FLASK – Project

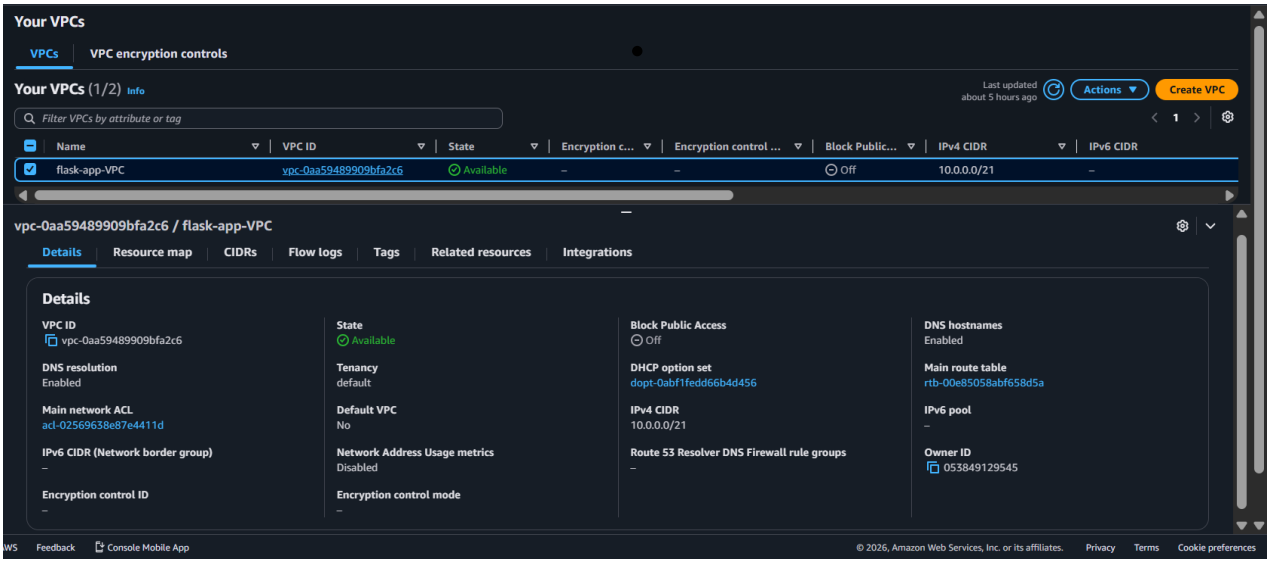
Region – N. virginia

Zone – 1a & 1b

Task – 1 (VPC, Subnet and Route tables configurations)

Name – flask -app-VPC

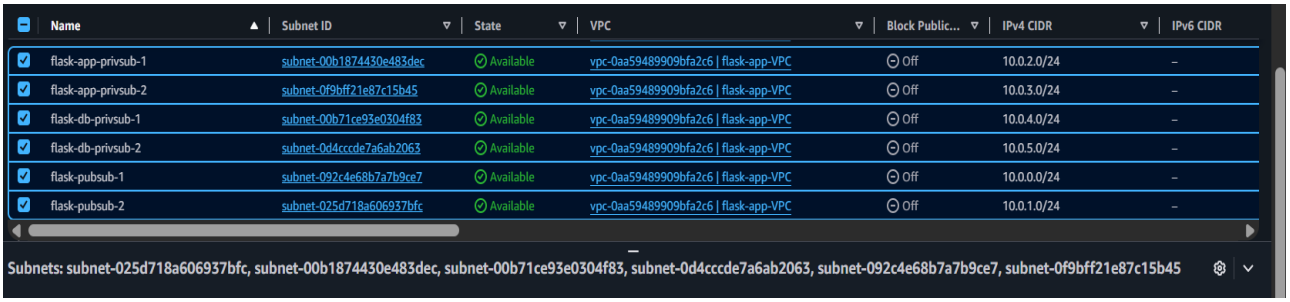
CIDR – 10.0.0.0/21



Total 6 Subnet

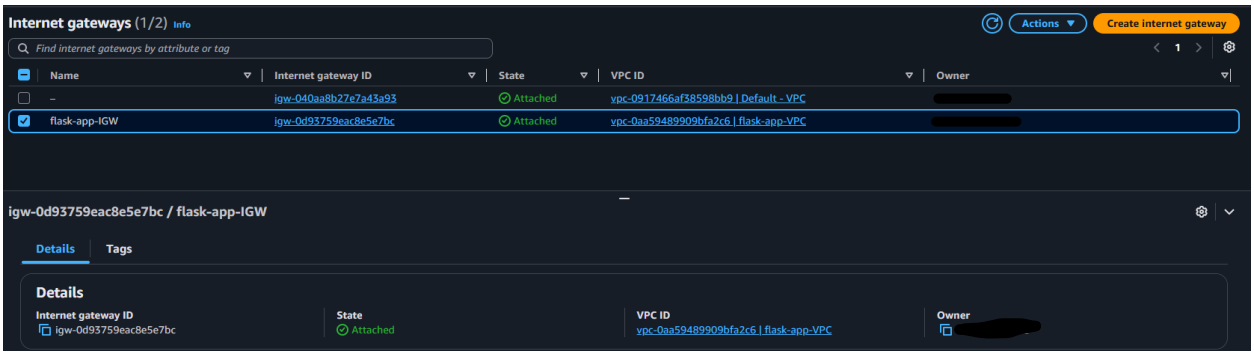
Subnet CIDR -- /24

Subnet Type	Quantity	Name	Details
Public (Web Tier)	2	flask-pubsub-1&2	10.0.0.0 &10.0.1.0
Private (Application Tier)	2	flask-app-privsub-1&2	10.0.2.0 &10.0.3.0
Private (Database Tier)	2	flask-db-privsub-1&2	10.0.4.0 &10.0.5.0

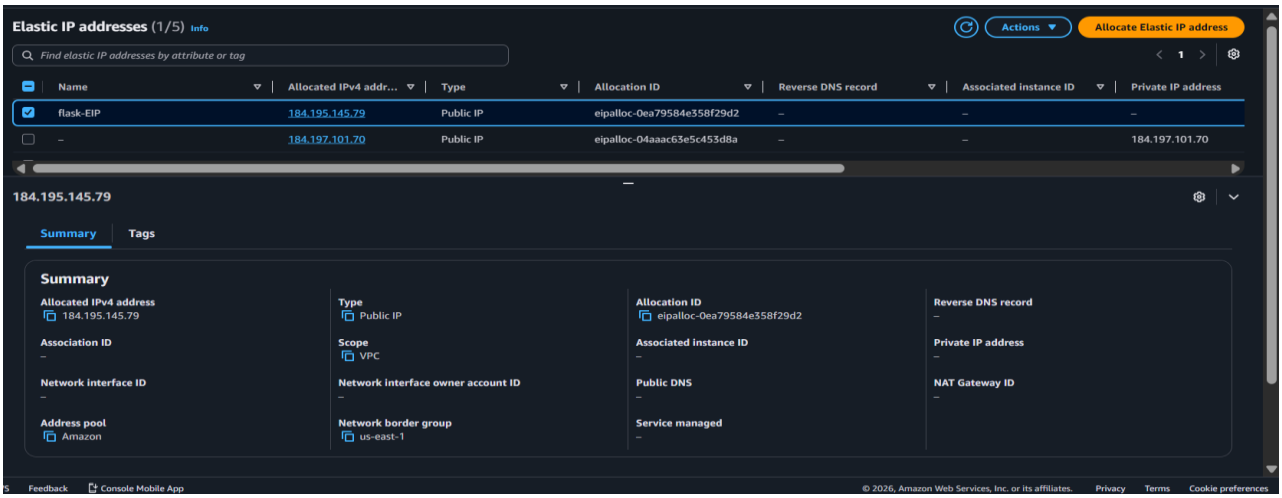


Internet gateway -- IGW

Name – flask-app-IGW

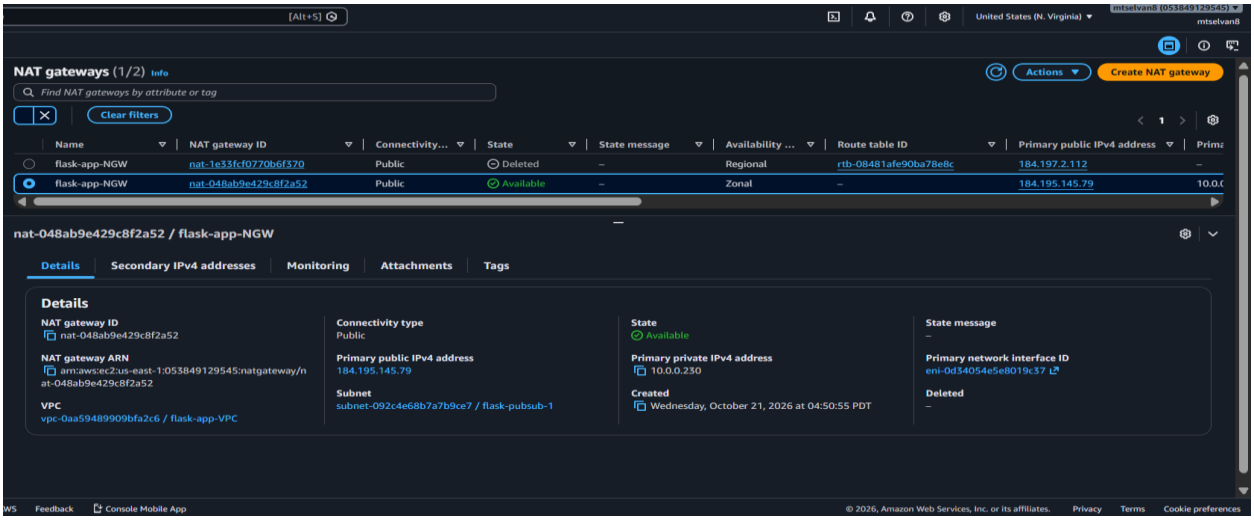


Elastic IP



NAT Gateway

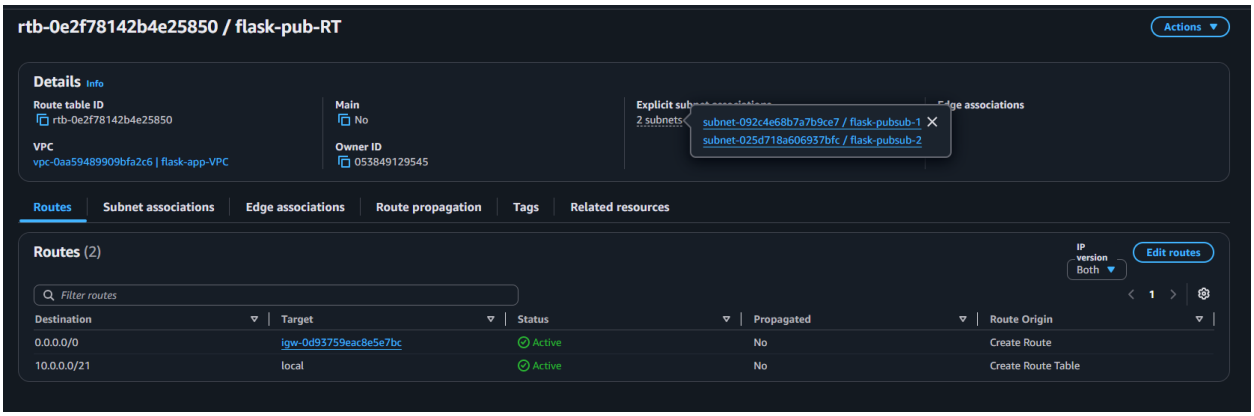
Name -- flask-app-NGW



Route Tables

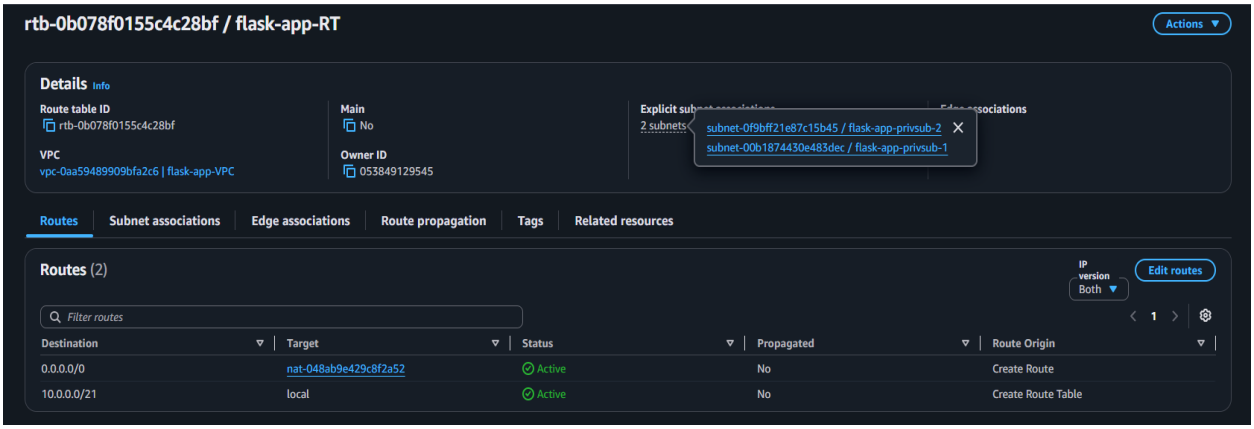
Public route table

Name -- flask-pub-RT



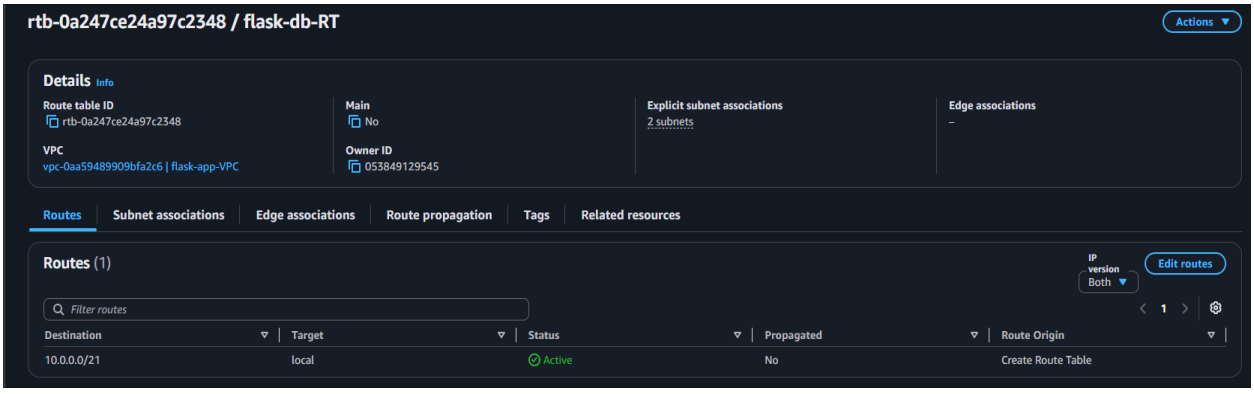
Private App route table

Name -- flask-app-RT



Private DB route table

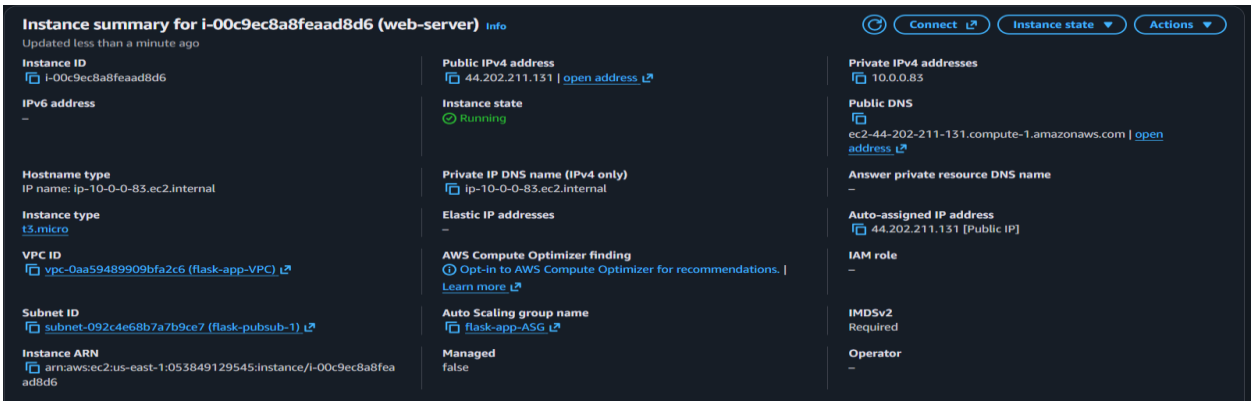
Name -- flask-db-RT



Task – 2,3,4&5

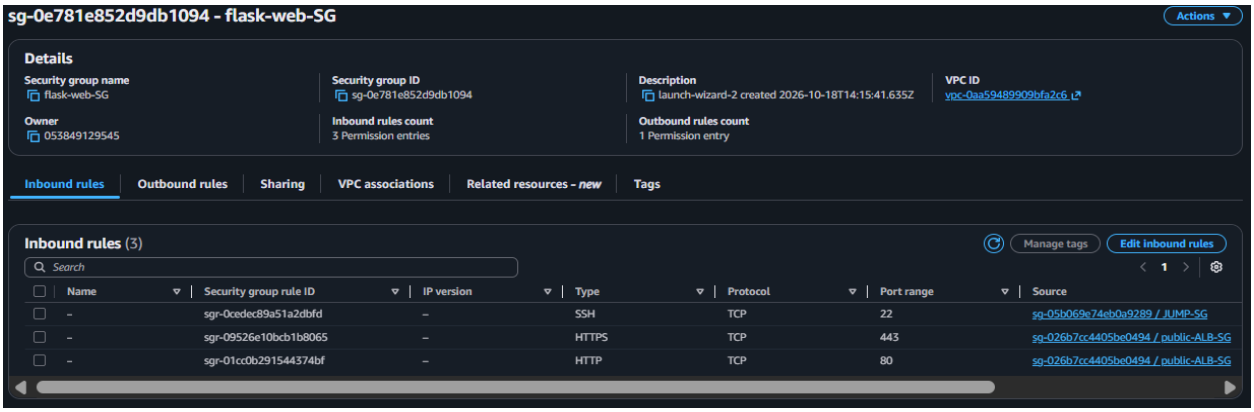
Servers, DB deployment and Security Group configuration

Name – web-server

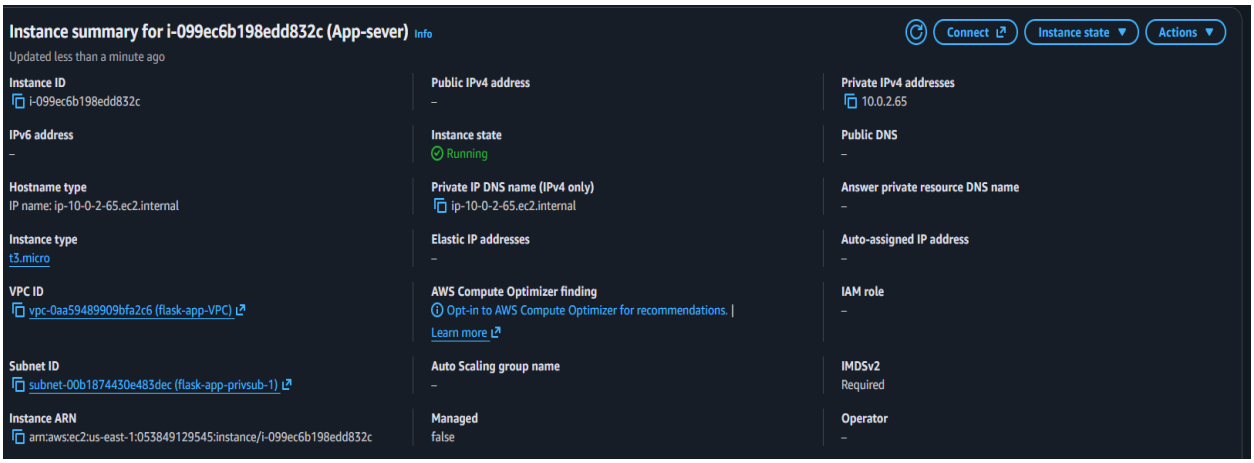


Web-server security group

Name – flask-web-SG **|| HTTP 80, HTTPS 443 – Source (Public-ALB-SG) & ssh 22 -- Source (JUMP-SG)**

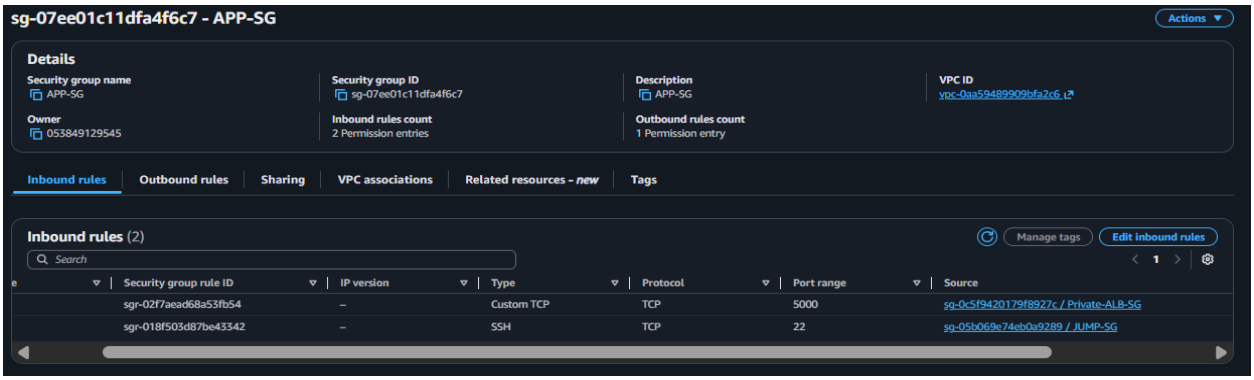


Name – App-server



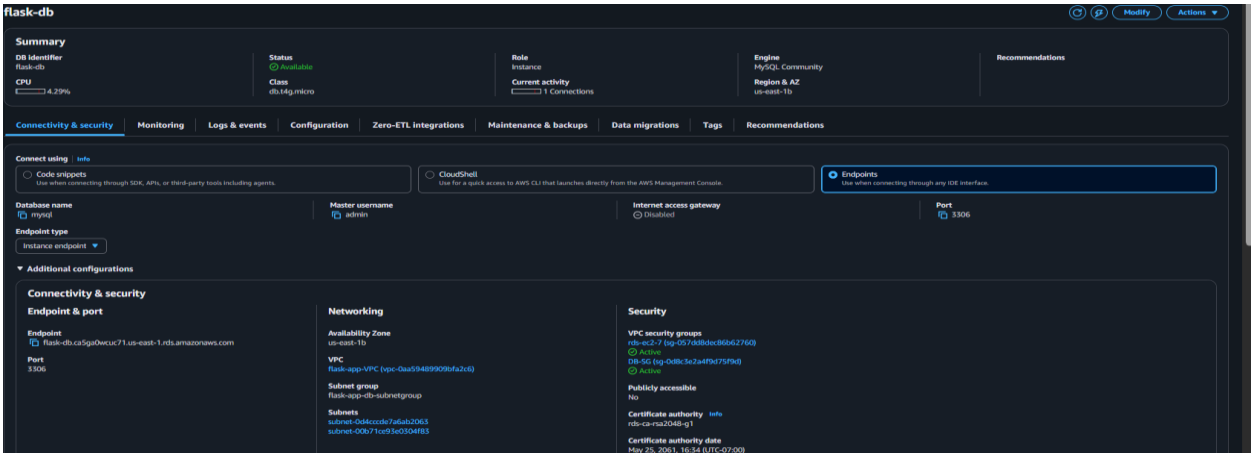
APP security group

Name – APP-SG **|| Custom TCP 5000 – Source (Private-ALB-SG) & ssh 22 -- Source (JUMP-SG)**



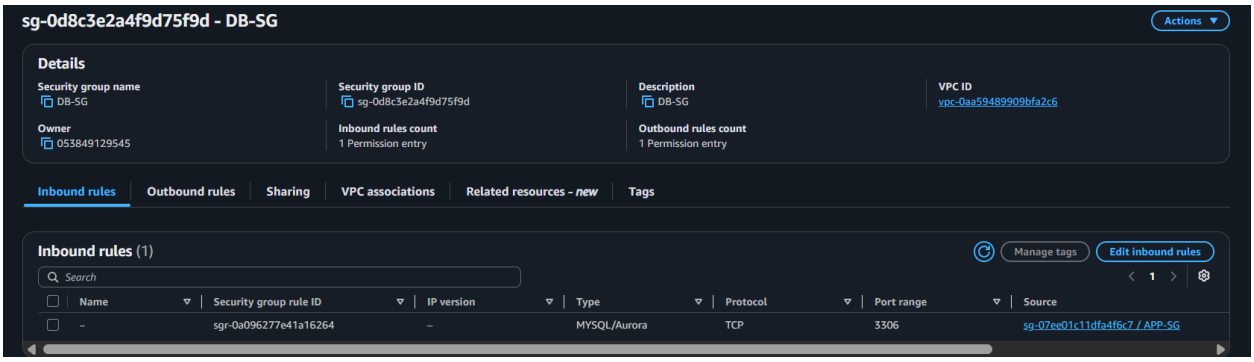
DB Instance

Name – flask-db



DB security group

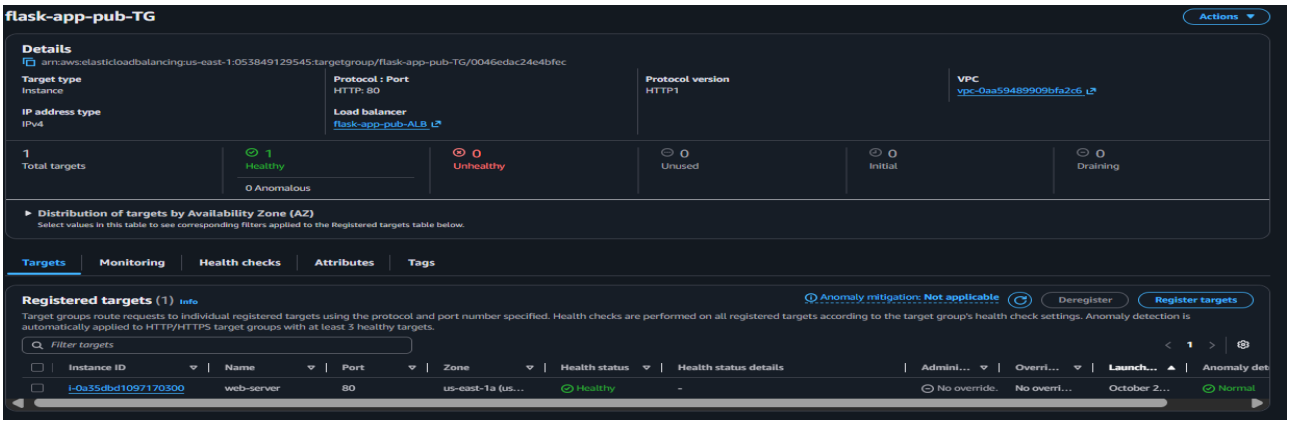
Name – DB-SG **|| MYSQL 3306 – Source (APP-SG)**



Task -6&7 – Load balancer and Target group configuration

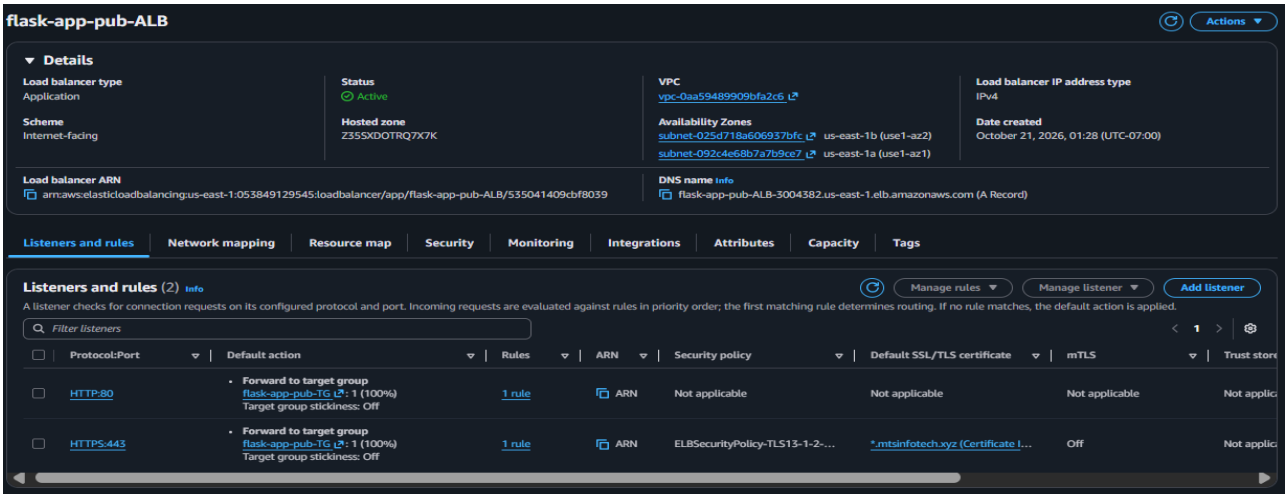
Target group for Internet facing ALB

Name –flask-app-pub-TG



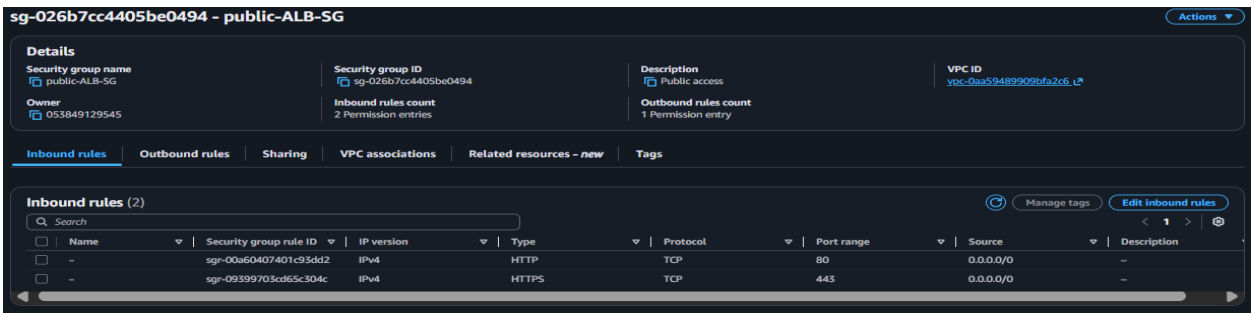
Internet facing ALB

Name -- flask-app-pub-ALB



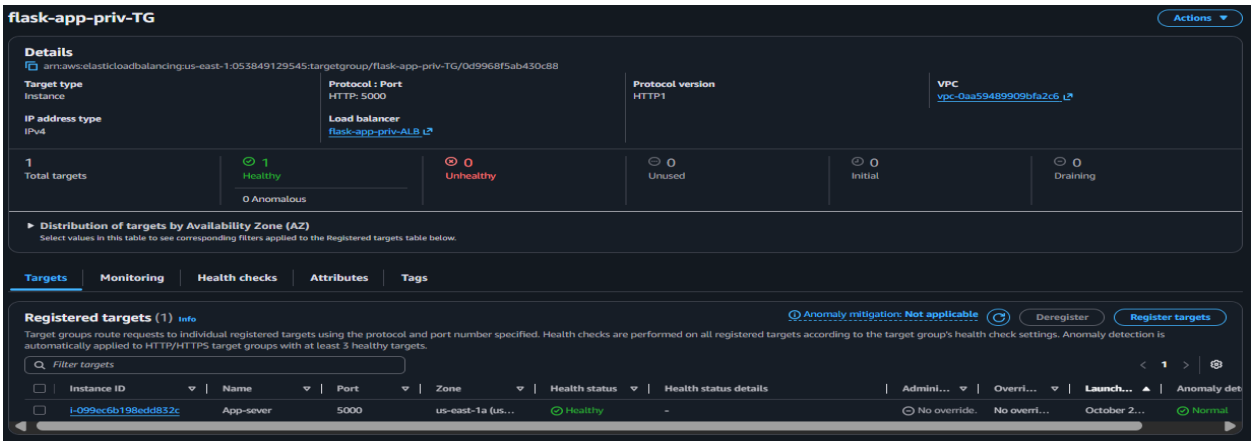
ALB security group

Name – Public-ALB-SG **|| HTTP 80, HTTPS 443 – Source (0.0.0.0/0)**



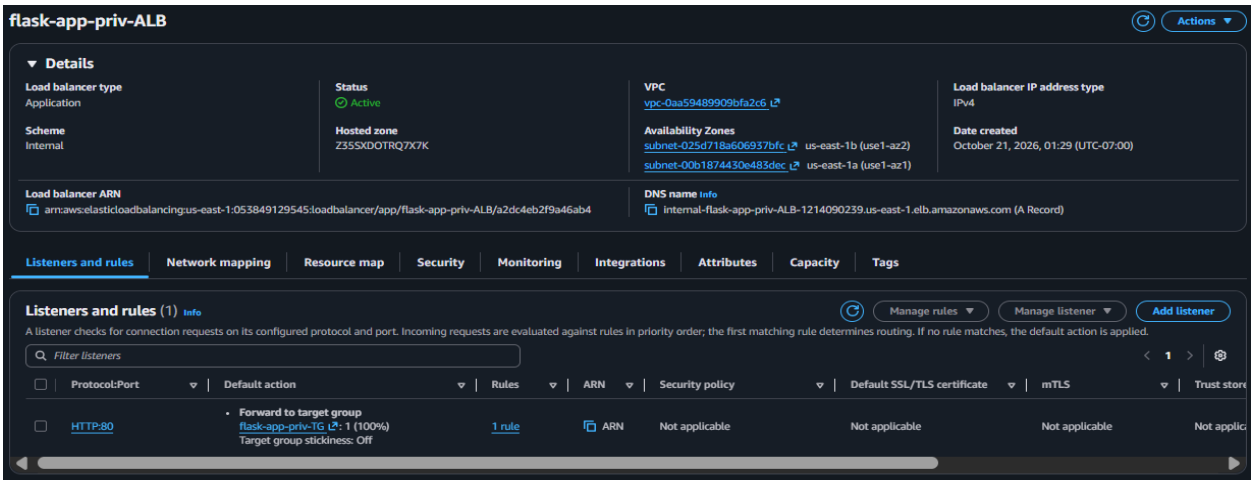
Target group for Internal ALB

Name –flask-app-priv-TG



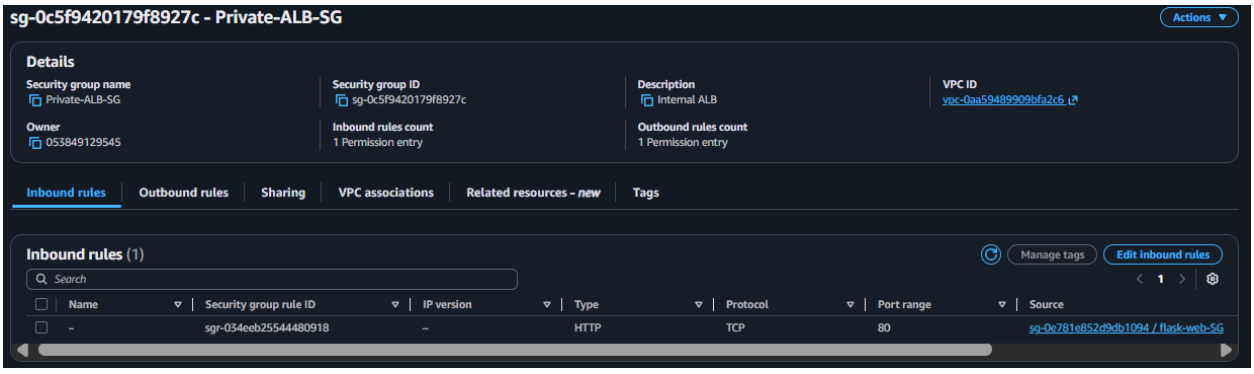
Private ALB

Name -- flask-app-Priv-ALB



ALB security group

Name – Private-ALB-SG **||HTTP 80 – Source (flask-web-SG)**



Task – 8 and 9 –nginx reverse proxy configuration and integrate the App to RDS

Verifying connectivity from web tier to App tier

```
[root@ip-10-0-0-83 ~]# curl  http://internal-flask-app-priv-ALB-1214090239.us-east-1.elb.amazonaws.com/
{"status":"Flask is running"}
[root@ip-10-0-0-83 ~]#
```

Verifying connectivity from App tier to DB tier

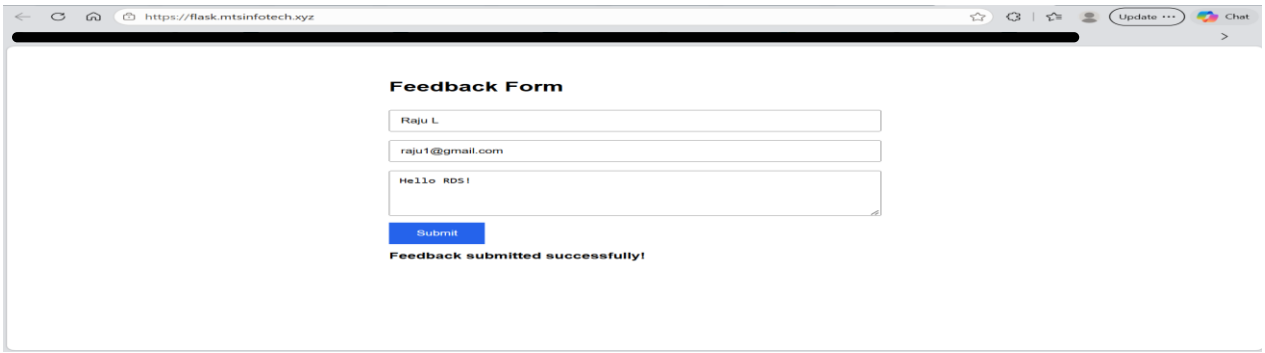
```
root@ip-10-0-2-65:~# nc -zv flask-db.ca5ga0wcuc71.us-east-1.rds.amazonaws.com 3306
Connection to flask-db.ca5ga0wcuc71.us-east-1.rds.amazonaws.com (10.0.5.143) 3306 port [tcp/mysql] succeeded!
root@ip-10-0-2-65:~#
```

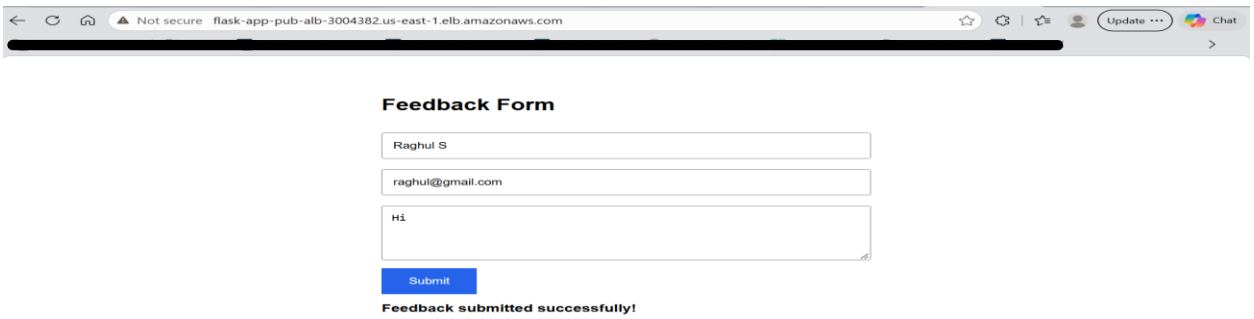
Task -10 – Verifying Web Application

Verifying URLs

Public ALB --- flask-app-pub-ALB-3004382.us-east-1.elb.amazonaws.com

URL -- https://flask.mtsinfotech.xyz





After Feedback form submission, Verifying DB

```
root@ip-10-0-2-65:~# mysql -h flask-db.ca5ga0wcuc71.us-east-1.rds.amazonaws.com -P 3306 -u admin -p --ssl-mode=VERIFY_IDENTITY --ssl-ca=./global-bundle.pem
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 256
Server version: 8.4.11 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| feedbackdb         |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.00 sec)
```

```
aws  Search [Alt+S] Ask Amazon Q

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| feedbackdb         |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.00 sec)

mysql> USE feedbackdb;
Database changed
mysql>
```

```
aws  Search [Alt+S] Ask Amazon Q  United States (N. Virginia)

mysql> SELECT * FROM feedback;
+----+----------------+---------------------+------------+---------------------+
| id | name           | email               | feedback   | created             |
+----+----------------+---------------------+------------+---------------------+
|  1 | Test           | t@t.com             | Hello RDS! | 2026-10-21 07:48:48 |
|  2 | Tamil Selvan M | mtselvan8@gmail.com | Ok         | 2026-10-21 09:41:20 |
+----+----------------+---------------------+------------+---------------------+
2 rows in set (0.00 sec)

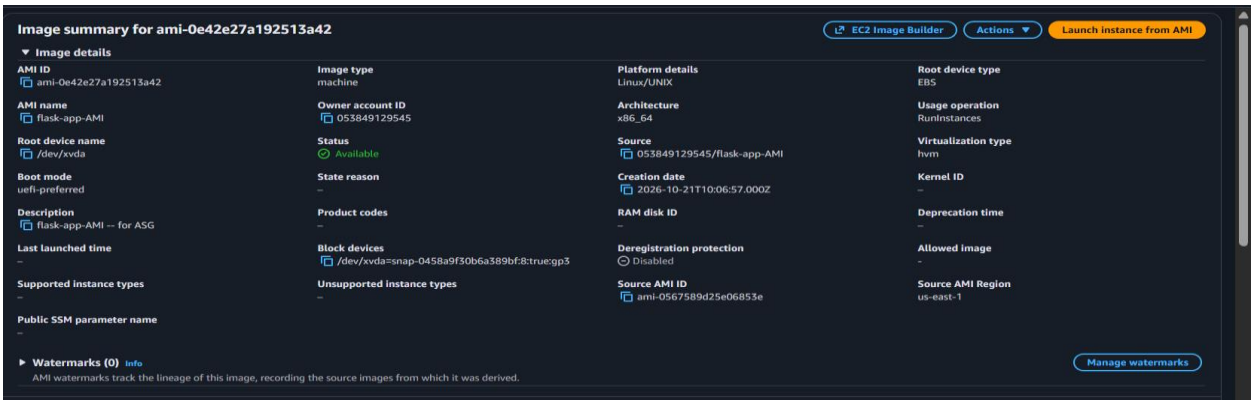
mysql> SELECT * FROM feedback;
+----+----------------+---------------------+------------+---------------------+
| id | name           | email               | feedback   | created             |
+----+----------------+---------------------+------------+---------------------+
|  1 | Test           | t@t.com             | Hello RDS! | 2026-10-21 07:48:48 |
|  2 | Tamil Selvan M | mtselvan8@gmail.com | Ok         | 2026-10-21 09:41:20 |
|  7 | Raju L         | raju1@gmail.com     | Hello RDS! | 2026-10-21 14:18:10 |
+----+----------------+---------------------+------------+---------------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM feedback;
+----+----------------+---------------------+------------+---------------------+
| id | name           | email               | feedback   | created             |
+----+----------------+---------------------+------------+---------------------+
|  1 | Test           | t@t.com             | Hello RDS! | 2026-10-21 07:48:48 |
|  2 | Tamil Selvan M | mtselvan8@gmail.com | Ok         | 2026-10-21 09:41:20 |
|  7 | Raju L         | raju1@gmail.com     | Hello RDS! | 2026-10-21 14:18:10 |
|  8 | Raghul S       | raghul@gmail.com    | Hi         | 2026-10-21 14:21:54 |
+----+----------------+---------------------+------------+---------------------+
4 rows in set (0.00 sec)
```

Configuring Auto-scaling Group with web tier

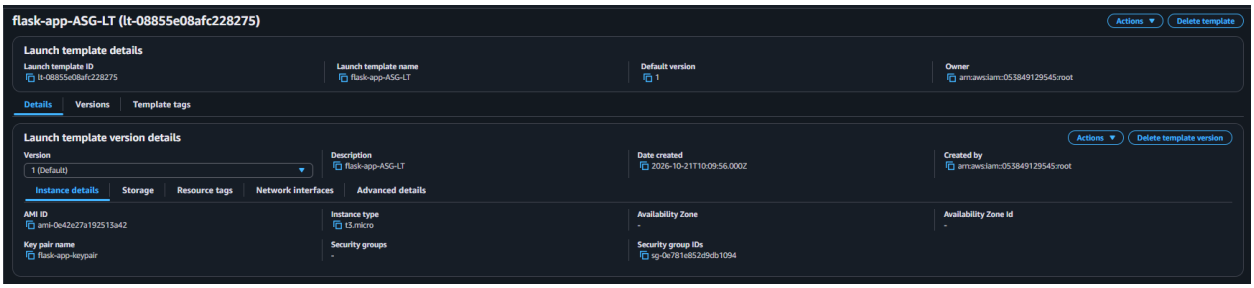
1 – AMI

Name – flask-app-AMI



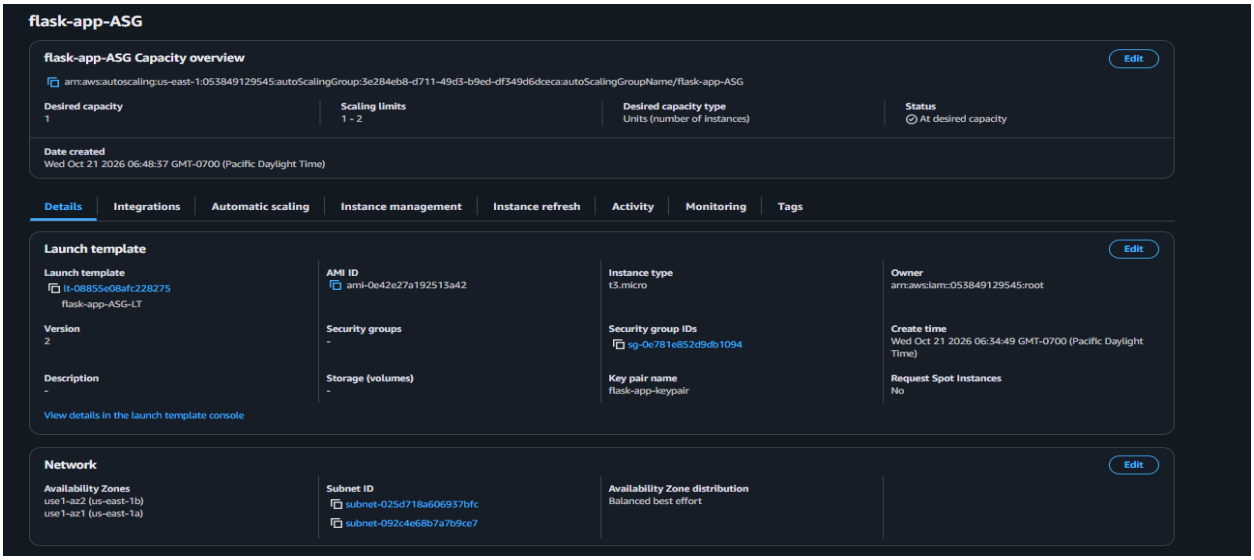
2-Launch Template

Name-flask-app-ASG-LT

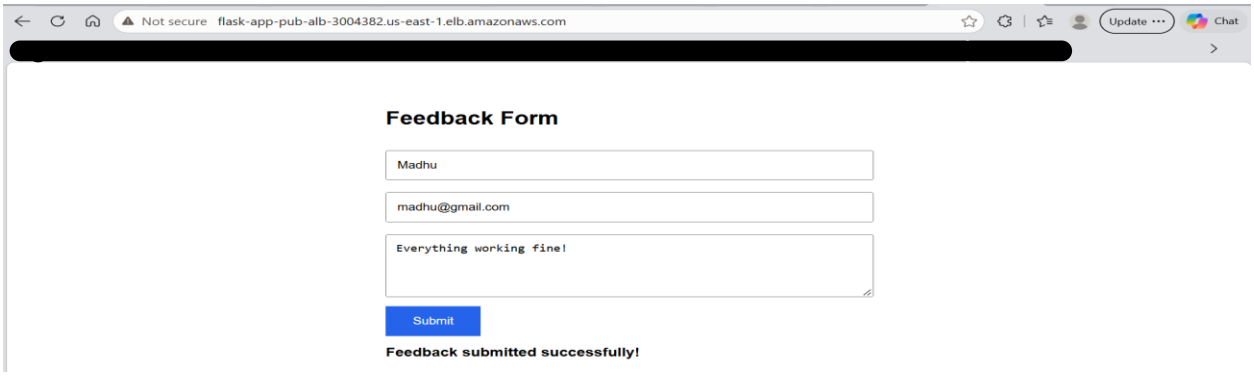


3 – Auto scaling group

Name – flask-app-ASG



Verifying after ASG configuration



Verifying DB

```
mysql> USE feedbackdb;
Database changed
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| feedbackdb         |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM feedback;
+----+---------------+--------------------+--------------------------+---------------------+
| id | name          | email              | feedback                 | created             |
+----+---------------+--------------------+--------------------------+---------------------+
|  1 | Test          | t@t.com            | Hello RDS!               | 2026-10-21 07:48:48 |
|  2 | Tamil Selvan M | mtselvan8@gmail.com | Ok                       | 2026-10-21 09:41:20 |
|  7 | Raju L        | raju1@gmail.com    | Hello RDS!               | 2026-10-21 14:18:10 |
|  8 | Raghul S      | raghul@gmail.com   | Hi                       | 2026-10-21 14:21:54 |
|  9 | Madhu         | madhu@gmail.com    | Everything working fine! | 2026-10-21 14:40:52 |
+----+---------------+--------------------+--------------------------+---------------------+
5 rows in set (0.00 sec)
```

Bastion server details

Name – JUMP-VM **|| SSH 22 – source (MY IP)**

